



Multiplication as Repeated Addition

Suppose there are 3 groups of 2 boys each.



How many boys are there in all?

Clearly, 6.

$$2 + 2 + 2 = 6.$$

Thus, 2 taken 3 times makes 6.

We write it as: $2 \times 3 = 6$.

We say: 2 multiplied by 3 gives 6.

$2 \times 3 = 6$ is a **multiplication fact**.

Now, take 4 groups of 3 gifts each.



How many gifts are there in all?

Clearly, 12.

$$3 + 3 + 3 + 3 = 12.$$

Thus, 3 taken 4 times makes 12.

We write it as: $3 \times 4 = 12$.

We say that 3 multiplied by 4 gives 12.

$3 \times 4 = 12$ is a multiplication fact.



Exercise 14

Write the multiplication fact for each of the following.

1.



$$\bigcirc \times \bigcirc = \bigcirc$$

2.



$$\bigcirc \times \bigcirc = \bigcirc$$

3.



$$\bigcirc \times \bigcirc = \bigcirc$$

4.



$$\bigcirc \times \bigcirc = \bigcirc$$

5.



$$\bigcirc \times \bigcirc = \bigcirc$$

6.



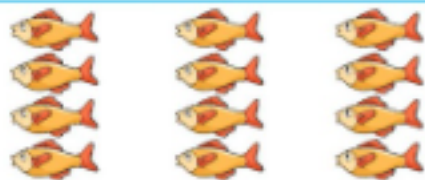
$$\bigcirc \times \bigcirc = \bigcirc$$

7.



$$\bigcirc \times \bigcirc = \bigcirc$$

8.



$$\bigcirc \times \bigcirc = \bigcirc$$

9.



$$\bigcirc \times \bigcirc = \bigcirc$$

10.



$$\bigcirc \times \bigcirc = \bigcirc$$

Multiplication Tables from 2 to 10

We have already learnt the multiplication tables from 2 to 10 in Class II.

Let us review these tables again.

Multiplication Table of 2

2	×	1	=	2
2	×	2	=	4
2	×	3	=	6
2	×	4	=	8
2	×	5	=	10
2	×	6	=	12
2	×	7	=	14
2	×	8	=	16
2	×	9	=	18
2	×	10	=	20

Multiplication Table of 3

3	×	1	=	3
3	×	2	=	6
3	×	3	=	9
3	×	4	=	12
3	×	5	=	15
3	×	6	=	18
3	×	7	=	21
3	×	8	=	24
3	×	9	=	27
3	×	10	=	30



Multiplication Table of 4

4	×	1	=	4
4	×	2	=	8
4	×	3	=	12
4	×	4	=	16
4	×	5	=	20
4	×	6	=	24
4	×	7	=	28
4	×	8	=	32
4	×	9	=	36
4	×	10	=	40

Multiplication Table of 5

5	×	1	=	5
5	×	2	=	10
5	×	3	=	15
5	×	4	=	20
5	×	5	=	25
5	×	6	=	30
5	×	7	=	35
5	×	8	=	40
5	×	9	=	45
5	×	10	=	50

Multiplication Table of 6

6	×	1	=	6
6	×	2	=	12
6	×	3	=	18
6	×	4	=	24
6	×	5	=	30
6	×	6	=	36
6	×	7	=	42
6	×	8	=	48
6	×	9	=	54
6	×	10	=	60



Multiplication Table of 7

7	×	1	=	7
7	×	2	=	14
7	×	3	=	21
7	×	4	=	28
7	×	5	=	35
7	×	6	=	42
7	×	7	=	49
7	×	8	=	56
7	×	9	=	63
7	×	10	=	70



Multiplication Table of 8

8	×	1	=	8
8	×	2	=	16
8	×	3	=	24
8	×	4	=	32
8	×	5	=	40
8	×	6	=	48
8	×	7	=	56
8	×	8	=	64
8	×	9	=	72
8	×	10	=	80

Multiplication Table of 9

9	×	1	=	9
9	×	2	=	18
9	×	3	=	27
9	×	4	=	36
9	×	5	=	45
9	×	6	=	54
9	×	7	=	63
9	×	8	=	72
9	×	9	=	81
9	×	10	=	90

Multiplication Table of 10

10	×	1	=	10
10	×	2	=	20
10	×	3	=	30
10	×	4	=	40
10	×	5	=	50
10	×	6	=	60
10	×	7	=	70
10	×	8	=	80
10	×	9	=	90
10	×	10	=	100



Exercise 15

Fill in the missing numerals.

- $2 \times 7 = \underline{\quad}$
- $3 \times 9 = \underline{\quad}$
- $4 \times 6 = \underline{\quad}$
- $3 \times 6 = \underline{\quad}$
- $6 \times 2 = \underline{\quad}$
- $8 \times 3 = \underline{\quad}$
- $5 \times 5 = \underline{\quad}$
- $7 \times 8 = \underline{\quad}$
- $6 \times 6 = \underline{\quad}$
- $8 \times 6 = \underline{\quad}$
- $6 \times 9 = \underline{\quad}$
- $9 \times 7 = \underline{\quad}$
- $5 \times 7 = \underline{\quad}$
- $7 \times 7 = \underline{\quad}$
- $8 \times 8 = \underline{\quad}$

Complete the grid by multiplying the numbers.

×	1	2	3	4	5	6	7	8	9	10
1	1									
2				8						
3								24		
4			12							
5							35			
6					30					
7									63	
8		16								
9						54				
10										100

Multiplication of a 2-digit Number by a 1-digit Number

Example: Multiply 23 by 2.

Solution:

T	O
2	3
×	2
4	6



Explanation: We proceed as follows:

Step 1: Arrange the numbers in column form as shown above.

Step 2: Multiply 3 ones by 2.

3 ones $\times$ 2 = 6 ones. Write 6 under ones column.

Step 3: Multiply 2 tens by 2.

2 tens $\times$ 2 = 4 tens. Write 4 under tens column.

So, $23 \times 2 = 46$.



Exercise 16

Multiply:

- | 1. | <table border="1"><thead><tr><th>T</th><th>O</th></tr></thead><tbody><tr><td>1</td><td>1</td></tr><tr><td>×</td><td>7</td></tr><tr><td colspan="2"><hr/></td></tr><tr><td colspan="2"><hr/></td></tr></tbody></table> | T | O | 1 | 1 | × | 7 | <hr/> | | <hr/> | | 2. | <table border="1"><thead><tr><th>T</th><th>O</th></tr></thead><tbody><tr><td>2</td><td>1</td></tr><tr><td>×</td><td>4</td></tr><tr><td colspan="2"><hr/></td></tr><tr><td colspan="2"><hr/></td></tr></tbody></table> | T | O | 2 | 1 | × | 4 | <hr/> | | <hr/> | | 3. | <table border="1"><thead><tr><th>T</th><th>O</th></tr></thead><tbody><tr><td>3</td><td>3</td></tr><tr><td>×</td><td>2</td></tr><tr><td colspan="2"><hr/></td></tr><tr><td colspan="2"><hr/></td></tr></tbody></table> | T | O | 3 | 3 | × | 2 | <hr/> | | <hr/> | | 4. | <table border="1"><thead><tr><th>T</th><th>O</th></tr></thead><tbody><tr><td>3</td><td>2</td></tr><tr><td>×</td><td>3</td></tr><tr><td colspan="2"><hr/></td></tr><tr><td colspan="2"><hr/></td></tr></tbody></table> | T | O | 3 | 2 | × | 3 | <hr/> | | <hr/> | | 5. | <table border="1"><thead><tr><th>T</th><th>O</th></tr></thead><tbody><tr><td>1</td><td>2</td></tr><tr><td>×</td><td>4</td></tr><tr><td colspan="2"><hr/></td></tr><tr><td colspan="2"><hr/></td></tr></tbody></table> | T | O | 1 | 2 | × | 4 | <hr/> | | <hr/> | |
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| 3 | 9 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
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Multiplication with Carrying

Example 1: Multiply 26 by 3.

Solution:

T	O
2	6
×	3
7	8

Explanation:

We proceed stepwise as follows:

Step 1: Write the numbers in column form.

Step 2: **Multiplying ones:** $6 \text{ ones} \times 3 = 18 \text{ ones}$
 $= 1 \text{ ten} + 8 \text{ ones}.$

Write 8 under ones column and carry over 1 to tens column.

Step 3: **Multiplying tens:** $2 \text{ tens} \times 3 = 6 \text{ tens}$

$6 \text{ tens} + 1 \text{ ten (carried over)} = 7 \text{ tens}$

Write 7 under tens column.

Thus, $26 \times 3 = 78$.

Example 2: Multiply 68 by 4.

Solution:

H	T	O
2	6	8
	×	4
2	7	2

Explanation:

Step 1: Write the numbers in column form.

Step 2: **Multiplying ones:** $8 \text{ ones} \times 4 = 32 \text{ ones}$
 $= 3 \text{ tens} + 2 \text{ ones}.$

Write 2 under ones column and carry over 3 to tens column.

Step 3: **Multiplying tens:** $6 \text{ tens} \times 4 = 24 \text{ tens}$

$24 \text{ tens} + 3 \text{ tens (carried over)} = 27 \text{ tens} = 2 \text{ hundreds} + 7 \text{ tens}.$

Write 7 under tens column and 2 under hundreds column.

Thus, $68 \times 4 = 272$.





Exercise 17

Multiply:

1.

T	O
2	3
×	4

2.

T	O
4	7
×	2

3.

T	O
2	8
×	3

4.

T	O
1	9
×	4

5.

T	O
1	8
×	5

6.

T	O
1	4
×	6

7.

T	O
4	9
×	2

8.

T	O
2	7
×	2

9.

T	O
1	2
×	8

10.

T	O
1	6
×	5

11.

T	O
3	7
×	2

12.

T	O
1	3
×	7

Multiply:

13.

H	T	O
	6	5
	×	3

14.

H	T	O
	3	4
	×	6

15.

H	T	O
	4	6
	×	5

16.

H	T	O
	7	8
	×	3

17.

H	T	O
	4	8
	×	4

18.

H	T	O
	5	9
	×	3

19.

H	T	O
	3	8
	×	5

20.

H	T	O
	6	3
	×	6

21.

H	T	O
	5	4
	×	4

22.

H	T	O
	4	3
	×	5

23.

H	T	O
	6	7
	×	3

24.

H	T	O
	7	5
	×	3

25.

H	T	O
	1	8
	×	8

26.

H	T	O
	1	7
	×	9

27.

H	T	O
	4	3
	×	6

28.

H	T	O
	2	8
	×	7

29.

H	T	O
	1	3
	×	9

30.

H	T	O
	2	4
	×	8

31.

H	T	O
	8	8
	×	7

32.

H	T	O
	4	9
	×	9



33.

H	T	O
	9	8
	×	8

34.

H	T	O
	9	4
	×	9



Multiplication Tables from 11 to 20

Multiplication Table of 11

11	×	1	=	11
11	×	2	=	22
11	×	3	=	33
11	×	4	=	44
11	×	5	=	55
11	×	6	=	66
11	×	7	=	77
11	×	8	=	88
11	×	9	=	99
11	×	10	=	110



Multiplication Table of 12

12	×	1	=	12
12	×	2	=	24
12	×	3	=	36
12	×	4	=	48
12	×	5	=	60
12	×	6	=	72
12	×	7	=	84
12	×	8	=	96
12	×	9	=	108
12	×	10	=	120

Multiplication Table of 13

13	×	1	=	13
13	×	2	=	26
13	×	3	=	39
13	×	4	=	52
13	×	5	=	65
13	×	6	=	78
13	×	7	=	91
13	×	8	=	104
13	×	9	=	117
13	×	10	=	130

Multiplication Table of 14

14	×	1	=	14
14	×	2	=	28
14	×	3	=	42
14	×	4	=	56
14	×	5	=	70
14	×	6	=	84
14	×	7	=	98
14	×	8	=	112
14	×	9	=	126
14	×	10	=	140



Multiplication Table of 15

15	×	1	=	15
15	×	2	=	30
15	×	3	=	45
15	×	4	=	60
15	×	5	=	75
15	×	6	=	90
15	×	7	=	105
15	×	8	=	120
15	×	9	=	135
15	×	10	=	150

Multiplication Table of 16

16	×	1	=	16
16	×	2	=	32
16	×	3	=	48
16	×	4	=	64
16	×	5	=	80
16	×	6	=	96
16	×	7	=	112
16	×	8	=	128
16	×	9	=	144
16	×	10	=	160

Multiplication Table of 17

17	×	1	=	17
17	×	2	=	34
17	×	3	=	51
17	×	4	=	68
17	×	5	=	85
17	×	6	=	102
17	×	7	=	119
17	×	8	=	136
17	×	9	=	153
17	×	10	=	170

Multiplication Table of 18

18	×	1	=	18
18	×	2	=	36
18	×	3	=	54
18	×	4	=	72
18	×	5	=	90
18	×	6	=	108
18	×	7	=	126
18	×	8	=	144
18	×	9	=	162
18	×	10	=	180



Multiplication Table of 19

19	×	1	=	19
19	×	2	=	38
19	×	3	=	57
19	×	4	=	76
19	×	5	=	95
19	×	6	=	114
19	×	7	=	133
19	×	8	=	152
19	×	9	=	171
19	×	10	=	190



Multiplication Table of 20

20	×	1	=	20
20	×	2	=	40
20	×	3	=	60
20	×	4	=	80
20	×	5	=	100
20	×	6	=	120
20	×	7	=	140
20	×	8	=	160
20	×	9	=	180
20	×	10	=	200



Exercise 18

Fill in the blanks.

- | | | |
|---------------------------|---------------------------|---------------------------|
| 1. $12 \times 5 =$ | 2. $13 \times 8 =$ | 3. $12 \times 7 =$ |
| 4. $14 \times 6 =$ | 5. $12 \times 9 =$ | 6. $11 \times 8 =$ |
| 7. $13 \times 7 =$ | 8. $14 \times 7 =$ | 9. $15 \times 8 =$ |
| 10. $16 \times 4 =$ | 11. $15 \times 9 =$ | 12. $14 \times 9 =$ |
| 13. $16 \times 7 =$ | 14. $18 \times 4 =$ | 15. $19 \times 5 =$ |
| 16. $17 \times 6 =$ | 17. $19 \times 7 =$ | 18. $16 \times 9 =$ |
| 19. $19 \times 8 =$ | 20. $18 \times 8 =$ | 21. $20 \times 7 =$ |

22. Look at the patterns and complete them.

- | | |
|----------------------|----------------------|
| (a) 5, 10, 15, | (b) 10, 20, 30, |
| (c) 3, 6, 9, | (d) 7, 14, 21, |
| (e) 9, 18, 27, | (f) 15, 30, 45, |
| (g) 20, 40, 60, | (h) 11, 22, 33, |
| (i) 12, 24, 36, | (j) 17, 34, 51, |

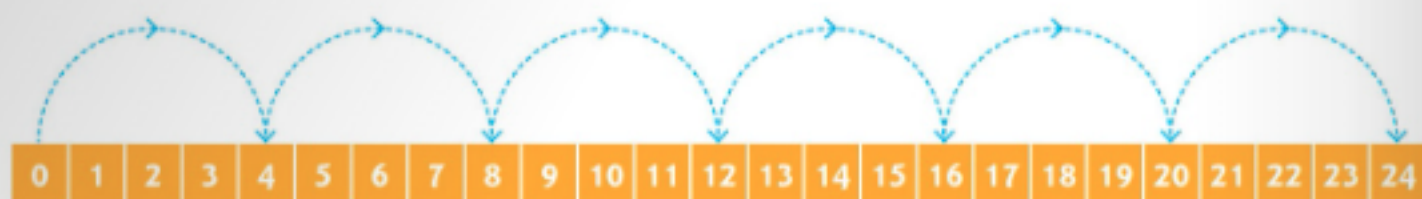


Complete the grid by multiplying the numbers in the uppermost row with the numbers in the leftmost column.

$\times$	11	12	13	14	15	16	17	18	19	20
1	11									
2					30					
3		36								
4				56						
5								90		
6			78							
7							119			
8										160
9									171	
10						160				

Skip Counting and Multiplication

Suppose a frog start from 0 and jumps 4 steps at a time on the number-strip as shown below.



Clearly,

- (i) after 1 jump it reaches 4;
 - (ii) after 2 jumps it reaches 8;
 - (iii) after 3 jumps it reaches 12;
 - (iv) after 4 jumps it reaches 16;
 - (v) after 5 jumps it reaches 20;
 - (vi) after 6 jumps it reaches 24;
 - (vii) after 7 jumps it reaches 28;
- and so on.

$$4 \times 1 = 4$$

$$4 \times 2 = 8$$

$$4 \times 3 = 12$$

$$4 \times 4 = 16$$

$$4 \times 5 = 20$$

$$4 \times 6 = 24$$

$$4 \times 7 = 28$$



Now, answer the following questions.

- (i) Starting from 0, Anu jumps 6 steps at a time. What number does she reach after 8 jumps?
- (ii) Starting from 0, Joe jumps 11 steps at a time. What number does he land on at the 9th jump?
- (iii) Starting from 0, Gauri jumps 8 steps at a time. What number does she reach after 7 jumps?
- (iv) Starting from 0, Adan jumps 12 steps at a time. What number does she reach after 15 jumps?
- (v) Starting from 0, a frog takes jumps of only 8. What would be the largest number that the frog will reach before crossing 100?
- (vi) Starting from 0, a frog takes jumps of only 7. What would be the least number that the frog will reach after crossing 50?
- (vii) Starting from 0, a frog makes 11 jumps of 9 steps each. How many odd numbers does it land on?
- (viii) A flower is placed on the number 16 on a number strip. It is required to start from 0 and skip equal steps to reach the flower. No, direct jumping to the flower is allowed. The one who reaches the flower in the smallest number of jumps wins. What skip jumping number will you choose to win?

Multiplication of 3-digit Numbers

In a multiplication sum,

- the number to be multiplied is called the **multiplicand**;
- the number by which, we multiply is called the **multiplier**; and
- the result of multiplication is called the **product**.

Thus, in $6 \times 5 = 30$, we have: Multiplicand = 6, Multiplier = 5 and Product = 30.

Multiplication by a 1-digit Number (Without Carrying)

Example 1: Multiply 312 by 3.

Solution:

H	T	O
3	1	2
	$\times$	3
9	3	6



Explanation: We proceed as follows:

Step 1: Multiply 2 ones by 3.

2 ones $\times$ 3 = 6 ones. Write 6 under ones column.

Step 2: Multiply 1 ten by 3.

1 ten $\times$ 3 = 3 tens. Write 3 under tens column.

Step 3: Multiply 3 hundreds by 3.

3 hundreds $\times$ 3 = 9 hundreds. Write 9 under hundreds column.

Thus, $312 \times 3 = 936$.

Multiplication by a 1-digit Number (With Carrying)

Example 2: Multiply 486 by 7.

Solution:

Th	H	T	O
(3)	(6)	(4)	
	4	8	6
		$\times$	7
3	4	0	2



Explanation: We proceed as follows:

Step 1:

Multiply 6 ones by 7.

$$6 \text{ ones} \times 7 = 42 \text{ ones}$$

$$= 4 \text{ tens} + 2 \text{ ones}$$

Write 2 under ones column. Carry over 4 tens.

Step 2:

Multiply 8 tens by 7.

$$8 \text{ tens} \times 7 = 56 \text{ tens}$$

$$\text{Now, } 56 \text{ tens} + 4 \text{ tens (carried over)} = 60 \text{ tens}$$

$$= 6 \text{ hundreds} + 0 \text{ tens.}$$

Write 0 under tens column. Carry over 6 hundreds.

Step 3:

Multiply 4 hundreds by 7.

$$4 \text{ hundreds} \times 7 = 28 \text{ hundreds}$$

$$\text{Now, } 28 \text{ hundreds} + 6 \text{ hundreds (carried over)}$$

$$= 34 \text{ hundreds}$$

$$= 3 \text{ thousands} + 4 \text{ hundreds.}$$

Write 4 under hundreds column and 3 under thousands column.

$$\text{Thus, } 486 \times 7 = 3402.$$



Exercise 19

Multiply:

1.

H	T	O
2	0	7
	$\times$	4

2.

H	T	O
1	3	2
	$\times$	3

3.

H	T	O
4	1	3
	$\times$	2

4.

H	T	O
2	1	0
	$\times$	4

5.

H	T	O
1	3	5
	$\times$	7

6.

H	T	O
1	6	9
	$\times$	4

7.

H	T	O
2	6	8
	$\times$	4

8.

H	T	O
1	8	6
	$\times$	5

9.

H	T	O
1	3	8
	$\times$	6

10.

H	T	O
1	9	4
	$\times$	5

11.

H	T	O
2	8	8
	$\times$	3

12.

H	T	O
3	1	6
	$\times$	3

13.

H	T	O
4	5	6
	$\times$	8

14.

H	T	O
6	0	8
	$\times$	5

15.

H	T	O
6	9	5
	$\times$	8

16.

H	T	O
7	8	2
	$\times$	5

17.

H	T	O
5	9	8
	$\times$	9

18.

H	T	O
7	9	3
	$\times$	7

19.

H	T	O
8	6	7
	$\times$	6

20.

H	T	O
9	8	6
	$\times$	7

21.

H	T	O
9	2	7
	$\times$	9

Multiplication by 10, 100, 1000

Example 1: Multiply 756 by 10. Record your observation.

Solution: $756 \times 10 = 756 \times 1 \text{ ten}$
 $= (756 \times 1) \text{ tens} = 756 \text{ tens} = 7560$
 $\therefore 756 \times 10 = 7560$

Observation: We get the product 756×10 by putting one zero to the right of 756.
 Thus, we have: $423 \times 10 = 4230$, $692 \times 10 = 6920$, $918 \times 10 = 9180$, etc.

Example 2: Multiply 69 by 100. Record your observation.

Solution: $69 \times 100 = 69 \times 1 \text{ hundred}$
 $= (69 \times 1) \text{ hundreds}$
 $= 69 \text{ hundreds} = 6900$
 $\therefore 69 \times 100 = 6900$

Observation: We get the product 69×100 by putting two zeros to the right of 69.
 Thus, we have: $4 \times 100 = 400$, $17 \times 100 = 1700$, $45 \times 100 = 4500$ etc.



Example 3: Multiply 8 by 1000. Record your observation.

Solution: $8 \times 1000 = 8 \times 1$ thousand
 $= (8 \times 1)$ thousands
 $= 8$ thousands $= 8000$
 $\therefore 8 \times 1000 = 8000.$

Observation: We get the product 8×1000 by putting three zeros to the right of 8.

Thus, we have: $3 \times 1000 = 3000$, $5 \times 1000 = 5000$ etc.

Now, carefully read the above observations. What do you conclude?

Conclusion: To multiply a given number by 10 or 100 or 1000, we put one or two or three zeros respectively to the right of the multiplicand and we get the product.

Multiplication by 20, 30, 40, ..., 90

Example 4: Find the products.

(a) 16×20

(b) 15×30

(c) 27×50

Solution: We have:

(a) $16 \times 20 = 16 \times 2$ tens
 $= (16 \times 2)$ tens
 $= 32$ tens $= 320.$
 $\therefore 16 \times 20 = 320.$

(b) $15 \times 30 = 15 \times 3$ tens
 $= (15 \times 3)$ tens
 $= 45$ tens $= 450.$
 $\therefore 15 \times 30 = 450.$

(c) $27 \times 50 = 27 \times 5$ tens
 $= (27 \times 5)$ tens
 $= 135$ tens $= 1350.$
 $\therefore 27 \times 50 = 1350.$



Conclusion: To multiply a given number by 20, 30, 40, ..., 90, we multiply the given number by 2, 3, 4,, 9 respectively and put a zero to the right of the product.

Multiplication by 200, 300, 400, ..., 900

Example 5: Find the products.

(a) 37×200

(b) 29×300

(c) 18×400

(d) 17×500

Solution: We have:

$$\begin{aligned} \text{(a) } 37 \times 200 &= 37 \times 2 \text{ hundreds} \\ &= (37 \times 2) \text{ hundreds} \\ &= 74 \text{ hundreds} = 7400. \\ \therefore 37 \times 200 &= 7400. \end{aligned}$$

$$\begin{aligned} \text{(b) } 29 \times 300 &= 29 \times 3 \text{ hundreds} \\ &= (29 \times 3) \text{ hundreds} \\ &= 87 \text{ hundreds} = 8700. \\ \therefore 29 \times 300 &= 8700. \end{aligned}$$

$$\begin{aligned} \text{(c) } 18 \times 400 &= 18 \times 4 \text{ hundreds} \\ &= (18 \times 4) \text{ hundreds} \\ &= 72 \text{ hundreds} = 7200. \\ \therefore 18 \times 400 &= 7200. \end{aligned}$$

$$\begin{aligned} \text{(d) } 17 \times 500 &= 17 \times 5 \text{ hundreds} \\ &= (17 \times 5) \text{ hundreds} \\ &= 85 \text{ hundreds} = 8500. \\ \therefore 17 \times 500 &= 8500. \end{aligned}$$



Conclusion: To multiply a given number by 200, 300, 400,, 900, we multiply the given number by 2, 3, 4, 9 respectively and put two zeros to the right of the product.



Exercise 20

Find the product.

1. 75×10

2. 387×10

3. 689×10

4. 100×10

5. 23×100

6. 38×100

7. 72×100

8. 84×100

9. 60×100

10. 4×1000

11. 7×1000

12. 6×1000

13. 57×20

14. 145×20

15. 67×30

16. 74×40

17. 40×50

18. 53×50

19. 61×60

20. 19×70

21. 60×90

22. 47×90

23. 75×80

24. 38×200

25. 16×400

26. 17×400

27. 27×300

28. 14×600

29. 12×800

30. 11×900

Properties of Multiplication

I. Order Property:

We may multiply two numbers in either order, the product remains the same.

This is known as order property of multiplication of numbers.

For example:

$$3 \times 4 = 12 \text{ and } 4 \times 3 = 12. \text{ Therefore, } 3 \times 4 = 4 \times 3.$$

$$6 \times 8 = 48 \text{ and } 8 \times 6 = 48. \text{ Therefore, } 6 \times 8 = 8 \times 6.$$

II. Grouping Property:

The product of three numbers remains the same even, when the grouping of the numbers is changed.

For example:

$$(7 \times 4) \times 3 = 28 \times 3 = 84 \text{ and } 7 \times (4 \times 3) = 7 \times 12 = 84.$$

$$\therefore (7 \times 4) \times 3 = 7 \times (4 \times 3).$$

$$\text{Similarly, } (8 \times 5) \times 14 = 40 \times 14 = 560 \text{ and } 8 \times (5 \times 14) = 8 \times 70 = 560.$$

$$\therefore (8 \times 5) \times 14 = 8 \times (5 \times 14).$$

III. Multiplicative Property of 1:

The product of any number and 1 is the number itself.

Clearly, $8 \times 1 = 8$ and $15 \times 1 = 15$.

Example: Find the products.

(a) 259×1

(b) 376×1

Solution:

(a)

H	T	O
2	5	9
	$\times$	1
2	5	9

(b)

H	T	O
3	7	6
	$\times$	1
3	7	6

Thus, $259 \times 1 = 259$ and $376 \times 1 = 376$.

IV. Multiplicative Property of Zero:

The product of any number and 0 is zero.

Clearly, $7 \times 0 = 0$, $19 \times 0 = 0$, $135 \times 0 = 0$, etc.



V. Distributive Property of Multiplication Over Addition:

Consider: $2 \times (3 + 4)$ and $(2 \times 3) + (2 \times 4)$.

We have: $2 \times (3 + 4) = 2 \times 7 = 14$.

And, $(2 \times 3) + (2 \times 4) = 6 + 8 = 14$.

Thus, $2 \times (3 + 4) = (2 \times 3) + (2 \times 4)$.

Similarly, you may verify the following:

(a) $5 \times (10 + 3) = (5 \times 10) + (5 \times 3)$.

(b) $4 \times (20 + 5) = (4 \times 20) + (4 \times 5)$.

(c) $7 \times (50 + 8) = (7 \times 50) + (7 \times 8)$.



Exercise 21

Without actual multiplication fill in the following blanks.

1. $18 \times \dots = 23 \times 18$
2. $\dots \times 35 = 35 \times 19$
3. $67 \times 29 = 29 \times \dots$
4. $39 \times 84 = \dots \times 39$
5. $75 \times \dots = 36 \times 75$
6. $\dots \times 96 = 96 \times 27$
7. $183 \times \dots = 183$
8. $\dots \times 271 = 271$
9. $870 \times 1 = \dots$
10. $1 \times 986 = \dots$
11. $192 \times 0 = \dots$
12. $0 \times 263 = \dots$
13. $(5 \times 8) \times 9 = 5 \times (\dots \times 9)$
14. $7 \times (6 \times 3) = (7 \times \dots) \times 3$
15. $9 \times (8 \times 10) = (9 \times \dots) \times 10$
16. $3 \times (5 \times 7) = (3 \times 5) \times \dots$
17. $4 \times (5 + 7) = (4 \times 5) + (4 \times \dots)$
18. $5 \times (10 + 6) = (5 \times 10) + (5 \times \dots)$
19. $5 \times (20 + 3) = (5 \times \dots) + (5 \times 3)$
20. $6 \times (30 + 9) = (6 \times \dots) + (6 \times \dots)$

Simple Word Problems

Example 1: 4 sweets were given to each boy in a group of 8 boys. How many sweets were distributed in all?

Solution: Number of sweets given to 1 boy = 4
Number of sweets given to 8 boys = $(4 \times 8) = 32$
Hence, 32 sweets were distributed in all.



Example 2: The cost of one chocolate is ₹ 8. Find the cost of 10 such chocolates.

Solution: Cost of 1 chocolate = ₹ 8
Cost of 10 chocolates = ₹ $(8 \times 10) = ₹ 80$
Hence, the cost of 10 chocolates is ₹ 80.



Example 3: A box can hold 7 pens. How many pens can 4 boxes hold?

Solution: Number of pens 1 box can hold = 7
Number of pens 4 boxes can hold = $(7 \times 4) = 28$
Hence, 4 boxes can hold 28 pens.



Example 4: A car travels a distance of 60 kilometres in 1 hour. How many kilometres will it cover in 7 hours?

Solution: Distance covered by the car in 1 hour = 60 km
Distance covered by the car in 7 hours = (60×7) km
= 420 km



Hence, the distance covered by the car in 7 hours is 420 km.

Example 5: In a grove, there are 24 trees in each row. If there are 80 rows of trees in all, how many trees are there in the grove?

Solution: Number of trees in 1 row = 24
Number of trees in 80 rows = 24×80
= 24×8 tens
= (24×8) tens
= 192 tens = 1920

Hence, the total number of trees in the grove is 1920.





Exercise 22

Solve the following word problems.

1. The cost of a ball pen is ₹ 5. What is the cost of 8 such ball pens?
2. There are 7 days in a week. How many days are there in 9 weeks?
3. There were 6 dozen bananas in a basket. How many bananas were there in all?
4. The cost of a cap is ₹ 50. What is the cost of 6 such caps?
5. The cost of a hairband is ₹ 9. What is the cost of 60 such hairbands?
6. One basket of mangoes weighs 36 kg. What will be the total weight of 10 such baskets of mangoes?
7. There are 18 sections in a school. Each section has 50 students. How many students are there in the school?
8. There are 24 hours in a day. How many hours are there in 30 days?
9. A packet contains 32 candles. How many candles are there in 40 such packets?
10. 65 people can travel in a bus. How many people can travel in 6 such buses?
11. A basket contains 135 apples. How many apples are there in 20 such baskets?
12. A cart can carry 168 bags of cement. How many bags can 10 such carts carry?
13. A book contains 80 pages. If each page contains 37 lines, how many lines are there in the whole book?
14. An aeroplane can fly 300 kilometres in 1 hour. How many kilometres can it fly in 24 hours?
15. There are 500 sheets of paper in a pack. How many sheets are there in 14 such packs?
16. There are 60 minutes in an hour. How many minutes are there in 12 hours?



Multiplication by a 2-digit Number

We use the distributive property of multiplication to solve this type of problems. The following examples will make it clear.

Example 1: Multiply 78 by 45.

Solution:

$$\begin{aligned}78 \times 45 &= 78 \times (40 + 5) \\&= (78 \times 40) + (78 \times 5) \\&= 3120 + 390 \\&= 3510.\end{aligned}$$

In a short method, we can do it as under.

Th	H	T	O	
		7	8	
	×	4	5	
	3	9	0	← (78 × 5)
3	1	2	0	← (78 × 40)
3	5	1	0	← (78 × 45)

Hence, $78 \times 45 = 3510$.

Example 2: Multiply 284 by 29.

Solution: We may write, $29 = 20 + 9$.

Th	H	T	O	
	2	8	4	
	×	2	9	
2	5	5	6	← (284 × 9)
5	6	8	0	← (284 × 20)
8	2	3	6	← (284 × 29)

Hence, $284 \times 29 = 8236$.



First multiply by the ones and then multiply by tens.





Exercise 23

Multiply:

1.
$$\begin{array}{r} 27 \\ \times 18 \\ \hline \end{array}$$

2.
$$\begin{array}{r} 58 \\ \times 26 \\ \hline \end{array}$$

3.
$$\begin{array}{r} 69 \\ \times 32 \\ \hline \end{array}$$

4.
$$\begin{array}{r} 74 \\ \times 39 \\ \hline \end{array}$$

5.
$$\begin{array}{r} 82 \\ \times 46 \\ \hline \end{array}$$

6.
$$\begin{array}{r} 95 \\ \times 53 \\ \hline \end{array}$$

7.
$$\begin{array}{r} 78 \\ \times 68 \\ \hline \end{array}$$

8.
$$\begin{array}{r} 56 \\ \times 77 \\ \hline \end{array}$$

9.
$$\begin{array}{r} 99 \\ \times 88 \\ \hline \end{array}$$

10.
$$\begin{array}{r} 87 \\ \times 93 \\ \hline \end{array}$$

11.
$$\begin{array}{r} 105 \\ \times 57 \\ \hline \end{array}$$

12.
$$\begin{array}{r} 134 \\ \times 65 \\ \hline \end{array}$$

13.
$$\begin{array}{r} 217 \\ \times 46 \\ \hline \end{array}$$

14.
$$\begin{array}{r} 308 \\ \times 28 \\ \hline \end{array}$$

15.
$$\begin{array}{r} 283 \\ \times 25 \\ \hline \end{array}$$

16.
$$\begin{array}{r} 169 \\ \times 42 \\ \hline \end{array}$$

17.
$$\begin{array}{r} 209 \\ \times 34 \\ \hline \end{array}$$

18.
$$\begin{array}{r} 250 \\ \times 18 \\ \hline \end{array}$$

19.
$$\begin{array}{r} 364 \\ \times 19 \\ \hline \end{array}$$

20.
$$\begin{array}{r} 473 \\ \times 18 \\ \hline \end{array}$$

21.
$$\begin{array}{r} 247 \\ \times 34 \\ \hline \end{array}$$

22.
$$\begin{array}{r} 407 \\ \times 16 \\ \hline \end{array}$$

23.
$$\begin{array}{r} 304 \\ \times 23 \\ \hline \end{array}$$

24.
$$\begin{array}{r} 503 \\ \times 19 \\ \hline \end{array}$$

25.
$$\begin{array}{r} 385 \\ \times 24 \\ \hline \end{array}$$

26.
$$\begin{array}{r} 199 \\ \times 47 \\ \hline \end{array}$$

27.
$$\begin{array}{r} 289 \\ \times 28 \\ \hline \end{array}$$

28.
$$\begin{array}{r} 179 \\ \times 29 \\ \hline \end{array}$$

Word Problems

Example 1: A bus runs at a speed of 65 km per hour.



How far will it go in 38 hours?

Solution:

Distance covered by the bus in 1 hour = 65 km

Distance covered by the bus in 38 hours = (65×38) km
= 2470 km

Hence, the bus will go 2470 km in 38 hours.

Th	H	T	O
		6	5
	×	3	8
	5	2	0
1	9	5	0
2	4	7	0

Example 2: The cost of a racket is ₹196. What is the cost of 27 such rackets?

Solution:

Cost of 1 racket = ₹ 196

Cost of 27 rackets = ₹ (196×27)
= ₹ 5292

Hence, the cost of 27 rackets is ₹ 5292.



Th	H	T	O
	1	9	6
	×	2	7
1	3	7	2
3	9	2	0
5	2	9	2

Example 3: A book contains 256 pages. Each page has 35 lines. How many lines are there in all in the book?

Solution:

Number of lines in 1 page = 35

Number of lines in 256 pages = (256×35)
= 8960

So, there are 8960 lines in all in the book.



Th	H	T	O
	2	5	6
	×	3	5
1	2	8	0
7	6	8	0
8	9	6	0

Example 4: A bag of rice weighs 23 kg. What is the total weight of 364 bags of rice?

Solution:

Weight of 1 bag of rice = 23 kg

Weight of 364 bags of rice = (364×23) kg
= 8372 kg

Hence, the total weight of 364 bags is 8372 kg.



Th	H	T	O
	3	6	4
	×	2	3
1	0	9	2
7	2	8	0
8	3	7	2



Exercise 24

1. Shruti can type 75 words a minute. How many words can she type in 48 minutes?
2. A box contains 48 chalk sticks. How many chalk sticks are there in 73 such boxes?
3. There are 24 slices of bread in one loaf. How many slices are there in 65 loaves?
4. A carton contains 165 books. How many books will 54 such cartons contain?
5. The cost of a chair is ₹ 375. What is the cost of 18 such chairs?
6. One basket of tomatoes contains 268 tomatoes. How many tomatoes do 36 such baskets contain?
7. A packet contains 148 buttons. How many buttons are there in 64 such packets?
8. The price of an electric heater is ₹ 685. How much would it cost to buy one dozen such heaters?
9. A truck can carry 352 gunny bags. How many gunny bags can be carried by 28 trucks?
10. In a cinema hall, there are 145 rows of chairs. In each row, there are 68 chairs. How many chairs are there in the cinema hall?
11. In a grove, there are 48 rows of coconut trees. In each row, there are 189 trees. How many coconut trees are there in the grove?
12. There are 23 coaches in a train. Each coach contains 125 seats. How many people can have seats on the train?
13. 42 boys in a class planned a picnic. If each boy contributed ₹ 168, how much money was collected?
14. Raju delivers 157 newspapers each morning. How many newspapers does he deliver in 31 days?
15. A small scale industry produces 346 screws in an hour. How many screws will it produce in 23 hours?





Activity Time



Make 10 identical cards, one of each of the digits 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9. The teacher asks each child to pick up 3 cards at random. With the 3 digits so chosen, the student must be asked to form:

1. A multiplication sum between a 3-digit number and a 1-digit number to yield the greatest product.

Example: If the chosen digits are 1, 2 and 4, the required multiplication sum is

Th	H	T	O
	4	4	4
		×	4
1	7	7	6



2. A multiplication sum between a 3-digit number (of different digits) and a 1-digit number to yield the greatest product.

Example: If the chosen digits are 1, 2 and 4, the required multiplication sum is

Th	H	T	O
	4	2	1
		×	4
1	6	8	4



3. A multiplication sum between a 3-digit number and a 1-digit number to yield the smallest product.

Example: If the chosen digits are 1, 2 and 4, the required multiplication sum is

Th	H	T	O
	1	1	1
		×	1
1	1	1	



4. A multiplication sum between a 3-digit number (of different digits) and a 1-digit to yield the smallest product.

Example: If the chosen digits are 1, 2 and 4,
the required multiplication sum is

Th	H	T	O
	1	2	4
		×	1
	1	2	4



Remember that for the greatest product, both the multiplicand and the multiplier must be the greatest and for the smallest product, both should be the smallest.



Things to Remember

- In a multiplication sum,
 - the number to be multiplied is called the **multiplicand**;
 - the number by which, we multiply is called the **multiplier**;
 - the result obtained on multiplication is called the **product**.
- To multiply a given number by 10 or 100 or 1000, we put one or two or three zeros respectively to the right of the multiplicand to get the product.
- To multiply a given number by 20, 30, 40, ..., 90, we multiply the given number by 2, 3, 4, ..., 9 respectively and put one zero to the right of the product.
- To multiply a given number by 200, 300, 400, ..., 900, we multiply the given number by 2, 3, 4, ..., 9 respectively and put two zeros to the right of the product.
- The product of two numbers remains the same even when their order is changed. This is known as order property of multiplication of numbers.
- The product of three numbers remains the same even when the grouping of the numbers is changed. This is known as grouping property of multiplication of numbers.
- The product of any number and 1 is the number itself.
- The product of any number and zero is zero.
- Distributive Property of Multiplication over Addition:
 $2 \times (3 + 4) = (2 \times 3) + (2 \times 4)$

**QUESTION BAG 1****(Objective Type Questions)**

Tick (✓) the correct answer.

1. In $8 \times 5 = 40$, 40 is called the

(a) multiplicand ☐
(c) sum ☐

(b) multiplier ☐
(d) product ☐



2. $10 \times 100 = ?$

(a) 10 ☐ (b) 100 ☐ (c) 1000 ☐ (d) 10000 ☐

3. If $7 \times 11 \times 13 = 1001$, then what is $11 \times 7 \times 13$?

(a) 77 ☐ (b) 91 ☐ (c) 143 ☐ (d) 1001 ☐

4. Which of the following is equal to 106×40 ?

(a) $(100 \times 4) + (6 \times 4)$ ☐ (b) $(100 \times 4) + (6 \times 40)$ ☐
(c) $(100 + 40) \times (6 + 40)$ ☐ (d) $(100 \times 40) + (6 \times 40)$ ☐

5. There are 48 beads in a necklace. How many beads are there in 8 necklaces?

(a) 48×8 ☐ (b) $48 + 8$ ☐ (c) $48 + 8 \times 6$ ☐ (d) $40 \times 8 + 48$ ☐

6. The product of the greatest 3-digit number and the greatest 1-digit number is

(a) 999 ☐ (b) 1997 ☐ (c) 8991 ☐ (d) 9990 ☐

7. Tina bought the following box of pencils. If each pencil costs ₹ 3, how much did all the pencils cost?

(a) ₹ 16 ☐ (b) ₹ 18 ☐ (c) ₹ 24 ☐ (d) ₹ 30 ☐



8. Which word problem below could be represented by the multiplication fact $8 \times 3 = 24$?

(a) Sachin had 8 books. He bought 3 more books. How many books did he have in all? ☐
(b) Sachin bought 8 packets of 3 books each. How many books did he buy? ☐
(c) Sachin had 8 books. He gave away 3 of them. How many books is he left with? ☐
(d) Sachin had 8 books. He put the books in 3 stacks. How many books were there in each stack? ☐

9. The product of 50 and 50 is
 (a) 25 ☐ (b) 250 ☐ (c) 500 ☐ (d) 2500 ☐
10. The product of 122 and 5 is
 (a) 710 ☐ (b) 720 ☐ (c) 610 ☐ (d) 620 ☐
11. The ones digit of the product 82 and 54 is
 (a) 2 ☐ (b) 4 ☐ (c) 6 ☐ (d) 8 ☐
12. The product of 16 and 238 is equal to
 (a) 3808 ☐ (b) 3818 ☐ (c) 2808 ☐ (d) 2818 ☐
13. Which of the following multiplication facts is not correct?
 (a) $63 \times (27 \times 18) = (18 \times 63) \times 27$ ☐ (b) $96 \times 14 = 96 \times 10 \times 4$ ☐
 (c) $87 \times 98 = (80 \times 98) + (7 \times 98)$ ☐ (d) $65 \times 20 \times 5 = 65 \times 100$ ☐
14. Kewal's mother bought 8 cases of mineral water for a party. There were 24 bottles in each case. How many bottles did she buy altogether?
 (a) 132 ☐ (b) 172 ☐ (c) 184 ☐ (d) 192 ☐
15. In the multiplication example shown on the right, the value of A is
 (a) 2 ☐ (b) 3 ☐
 (c) 4 ☐ (d) 6 ☐
- | | | |
|---|----------|---|
| H | T | O |
| | A | 7 |
| | $\times$ | 4 |
| 1 | 8 | 8 |
16. Anna has 9 pizzas that are cut into 6 slices each. She herself ate 2 slices while her 12 friends ate 3 slices each. How many slices are left?
 (a) 16 ☐ (b) 18 ☐ (c) 24 ☐ (d) 27 ☐
17. A man bought 3 burgers and 2 sandwiches. Each burger costs ₹ 24 and each sandwich costs ₹ 18. How much money does he have to pay altogether?
 (a) ₹ 102 ☐ (b) ₹ 108 ☐ (c) ₹ 112 ☐ (d) ₹ 120 ☐
18. The product of the greatest 2-digit number ending with 1 and the smallest 2-digit number ending with 1 is
 (a) 900 ☐ (b) 910 ☐ (c) 999 ☐ (d) 1001 ☐

QUESTION BAG 2

1. Fill in the missing numerals:

(a) $9 \times 7 =$

(c) $12 \times 8 =$

(e) $13 \times 9 =$

(g) $18 \times 8 =$

(i) $19 \times 4 =$

(k) $17 \times 6 =$

(m) $18 \times 9 =$



(b) $11 \times 6 =$

(d) $8 \times 8 =$

(f) $14 \times 5 =$

(h) $6 \times 9 =$

(j) $15 \times 7 =$

(l) $16 \times 8 =$

2. Fill in the missing numerals:

(a) $9 \times$ $= 72$

(c) $7 \times$ $= 49$

(e) $\times 8 = 112$

(g) $\times 7 = 91$

(b) $\times 8 = 56$

(d) $12 \times$ $= 84$

(f) $19 \times$ $= 114$

(h) $\times 8 = 120$



3. Put $>$, $<$ or $=$ in the boxes:

(a) 9×8 12×6

(c) 8×8 7×9

(e) 20×5 10×10

(g) 12×3 6×6

(i) 18×5 15×8



(b) 7×7 6×8

(d) 6×9 13×4

(f) 19×6 18×8

(h) 9×50 15×30

(j) 16×6 12×8

4. Solve:

(a) 80×6

(b) 7×40

(c) 50×7

(d) 3×60

(e) 9×30

(f) 20×8

(g) 70×6

(h) 5×90

5. Fill in the blank boxes:

(a) $923 \times 1 = \boxed{}$

(b) $84 \times 132 \times 0 = \boxed{}$

(c) $876 \times 10 = \boxed{}$

(d) $239 \times \boxed{} = 2390$

(e) $177 \times 10 = \boxed{}$

(f) $81 \times 100 = \boxed{}$

(g) $1 \times 8796 = \boxed{}$

(h) $63 \times \boxed{} = 6300$

(i) $10 \times 613 = \boxed{}$

(j) $\boxed{} \times 100 = 7600$

(k) $90 \times \boxed{} = 9000$

(l) $999 \times \boxed{} = 0$

(m) $25 \times 3 = 75 \times \boxed{}$

**6. Fill in the blank boxes:**

(a) $64 \times 84 = 84 \times \boxed{}$

(b) $97 \times 41 = 41 \times \boxed{}$

(c) $83 \times 24 = \boxed{} \times 83$

(d) $719 \times 86 = \boxed{} \times 719$

(e) $\boxed{} \times 754 = 754 \times 46$

(f) $93 \times 5 \times 37 = 37 \times \boxed{} \times 5$

(g) $27 \times 16 \times \boxed{} = 12 \times 27 \times 16$

(h) $9 \times (10 + \boxed{}) = 9 \times 15$

(i) $6 \times 39 = 6 \times (\boxed{} + 9)$

(j) $7 \times 44 = \boxed{} \times (40 + 4)$

7. Fill in the blank boxes:

(a) $9 \times 30 = 9 \times 3 \times \boxed{}$

(b) $63 \times 40 = \boxed{} \times 4 \times 10$

(c) $88 \times 50 = \boxed{} \times \boxed{} \times 10$

(d) $54 \times \boxed{} = 54 \times 7 \times 10$

(e) $74 \times 900 = 74 \times 9 \times \boxed{}$

(f) $87 \times 300 = \boxed{} \times \boxed{} \times 100$

(g) $\boxed{} \times 400 = 96 \times 4 \times 100$

(h) $18 \times \boxed{} = 18 \times 8 \times 100$

8. Fill in the blanks:

- There are days in 5 weeks.
- There are months in 9 years.
- In 6 dozen oranges there are oranges.
- One car has 4 wheels. So, 16 cars have wheels.
- One flower has 6 petals. So, 18 flowers have petals.
- A grasshopper covers 16 cm in every jump. So, it covers..... cm in 5 jumps.
- If there are 10 bottles in one carton, there are bottles in 98 cartons.
- If Mr Roy has 336 notes of ₹ 10, he has with him ₹ in all.
- I throw 7 dice, all sixes. My total score is
- One book has 90 pages. So, 12 books have pages.
- In one class, there are 30 students. So, there are students in 19 classes.



9. Find the value of:

- | | | |
|---------------------|---------------------|---------------------|
| (a) 14×30 | (b) 15×60 | (c) 38×20 |
| (d) 96×50 | (e) 16×700 | (f) 18×500 |
| (g) 17×400 | (h) 12×800 | |



10. Fill in the missing digits.

(a)

H	T	O
	3	□
×	□	6
2	□	8
□	6	0
9	□	□

(b)

H	T	O
	□	9
×	1	8
3	9	□
□	□	0
□	□	□

(c)

H	T	O
	□	6
×	3	□
1	5	6
□	□	0
□	3	□

11. There are 54 grapes in one bunch. If Tanvy bought 23 bunches, how many grapes does she have altogether?
12. A train had 13 compartments. There were 72 passengers in each of them. How many passengers were there in the train?
13. A bus can carry 66 children. How many children can go to picnic in 28 buses?
14. 45 students of a class collected ₹ 125 each for the Prime Minister's Relief Fund. How much money did they collect altogether?
15. Julie works as a nurse in a hospital. She works 8 hours a day for 6 days a week. How many hours does she work in 20 weeks?
16. A large hospital has 287 rooms for patients. If each room has 4 beds, how many beds are there in all?
17. A fruit seller bought 3 baskets of 196 apples each and 3 baskets of 244 mangoes each to sell in the market. How many fruits did he buy in all?
18. **State whether each of the following statements is true or false.**
 - (a) The product of two numbers is always greater than their sum.
 - (b) The product of two odd numbers is an even number.
 - (c) The product of two even numbers is an even number.
 - (d) The product of an odd and an even number is an odd number.

